

Edwards Heart Valve Replacement - Magn Ease

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Biomedical Engineering 1

Group 13

Abstract

This article focuses on the challenges and advances made in prosthetic heart valve technology, particularly the contributions of companies such as Edwards Lifesciences and Medtronic. Despite the progress, concerns about long-term durability and calcification continue to motivate research efforts. Utilization of long noncoding RNA lncTSI as a potential therapeutic target for blocking CAVD and chemical treatments proposed by the Fraunhofer Institute to prevent calcium deposition are the most promising solutions. The article also highlights the U.S. Food and Drug Administration's (FDA) stringent regulatory pathway for class III medical devices, emphasizing the importance of robust clinical data for approval. Addressing these challenges is critical to advancing prosthetic heart valve technology, ensuring safety, and improving outcomes for patients with valve disease.

Background

Heart valve diseases involve either the malfunctioning or damage of one or more of the heart's valves, affecting the heart's ability to circulate blood efficiently. These conditions can lead to significant health issues, including heart failure, stroke, blood clots, and sudden cardiac death if left untreated. The prevalence of heart valve diseases is a growing concern globally due to aging populations and the increasing incidence of risk factors such as hypertension, diabetes, and coronary artery disease. According to data from the American Heart Association (AHA), heart valve disease affects an estimated 2.5% of the U.S. population, with the prevalence increasing significantly with age. The AHA also highlights that conditions such as aortic stenosis (a narrowing of the aortic valve opening) and mitral regurgitation (leakage of the mitral valve) are among the most common valve disorders leading to hospitalization and surgical intervention.

Risk factors contributing to heart valve diseases include age-related changes, infections affecting the heart, high blood pressure, high cholesterol, smoking, and obesity. These conditions can lead to the

deterioration or damage of heart valves, necessitating medical or surgical interventions to restore normal heart function and improve patient outcomes.

Heart valves play a crucial role in directing blood flow through the heart's four chambers and into the rest of the body. They ensure that blood moves forward through the heart and prevent it from flowing backward. When valves are damaged or diseased, they may not open or close properly, which can disrupt blood flow and force the heart to work harder to supply the body with the necessary blood. This added strain can lead to heart failure or other serious complications.

In cases where medication and lifestyle changes cannot adequately manage heart valve disease, valve repair or replacement may be necessary. The decision to repair or replace a valve depends on the severity of the disease, the specific valve affected, and the overall health of the patient. Valve replacement can be performed using mechanical valves, which are durable and long-lasting, or tissue valves, which may not last as long but do not usually require lifelong blood-thinning medication.

Competitive Landscape

The artificial heart valve competitive landscape is incredibly vast. Many companies make heart valves, each with many of their pros and cons. Some of these notable companies include Edwards Lifesciences, and Medtronic among many other companies. A lot of these companies are made specifically on heart valves and do not focus on many other sectors. In addition to this, the market for artificial heart valves is enormous. The market is valued at 7.52 billion dollars and is expected to double by 2028. (HTF Market Intelligence)⁵.



Figure 1
Edwards Lifesciences Heart Valve
(Edwards Lifesciences)²

Perhaps the most well-known heart valve is the one produced by Edwards Lifesciences. It is known for its innovative design and success. However, some cons of the valve are that it doesn't have an especially long lifespan at approximately 10 years, and issues such as calcification can

require patients to need more than one prosthetic heart valve. (Edwards Lifesciences)⁸.

Medtronic is also very well known for its heart valves. They produce some of the most advanced heart valves on the market. A very good pro with Medtronic heart valves is its long-term durability. These



Figure 2
Medtronic Heart Valve
(Medtronic)⁶

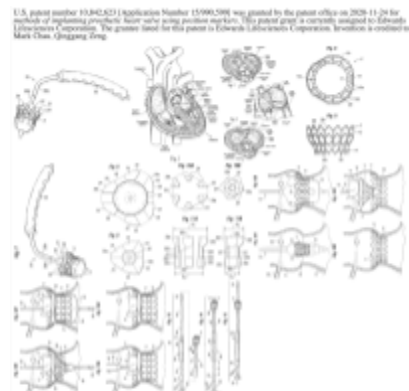
valves are known to last for 25 or more years without issue. This is why they are used more with younger patients who may have a longer life expectancy. However, there are some pretty big cons to consider with this heart valve. You will have to take blood-thinning medication for the rest of your life. This is because the mechanical heart valve can cause some blood clots, which if they get into the actual heart valve, may impact the function. (Medtronic)⁶.

Overall, many companies offer different kinds of valves that are specialized in different ways. Your lifestyle and age will make it easier to determine which one could be the best for you.

Description of Patent

Edwards Lifesciences Corporation has made significant contributions to the field of heart valve technology, as evidenced by its recent patents and patent applications. Included below is a description of each patent filed relating to the Prosthetic Heart valve.

- Sealing Member for Prosthetic Heart Valve (Patent number: 11883281): This patent introduces a prosthetic heart valve comprising an annular frame that is radially compressible and expandable, featuring a leaflet structure and an innovative outer sealing member. The



*This figure shows the inner
working of a heart valve*

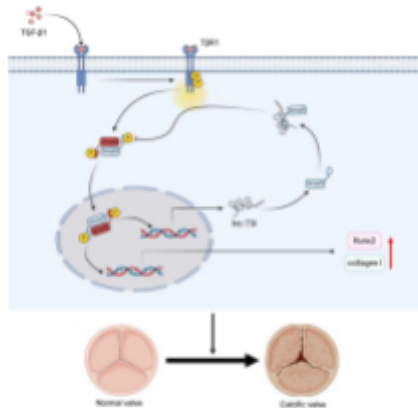
sealing member, designed to ensure a secure seal against surrounding tissue upon implantation, incorporates a mesh layer coupled with a pile layer of extending pile yarns, enhancing the valve's integration and performance within the heart.

- Assembly of Heart Valves and Intermediate Adapter Stent (Patent number: 11883282): The patent describes a unique approach to prosthetic heart valve implantation, involving a support frame that can be transitioned into an expanded form to accommodate or support an expandable prosthetic heart valve. This design allows for an initial rigid implantation configuration, which can later be expanded, providing versatility and adaptability in valve replacement procedures.
- Prosthetic Heart Valve (Patent number: 11883287): This application details a prosthetic heart valve that is both radially collapsible and expandable, equipped with a valve structure that includes multiple leaflets capable of transitioning between open and closed configurations. The frame supporting the valve structure is designed with an inflow end, an outflow end, and struts forming rows of cells, ensuring effective blood flow regulation through the leaflets' movements.

Proposed improvement(s)

Despite its promising features, such as the ThermaFix Process, which as written on their information page, “confronts both major calcium binding sites: residual glutaraldehydes and phospholipids (Edwards, n.d)⁸.” the Edwards Magna Ease aortic valve, faces limitations in long-term durability. These limitations stem from several factors such as limited follow-up data and mixed results in calcification. While initial studies show positive outcomes, the relatively short follow-up period (less than 10 years) makes it difficult to definitively assess the valve's long-term durability compared to other bioprosthetic options. The information page states that “no clinical data are available that evaluate the long-term impact of the Carpentier-Edwards ThermaFix issue process in patients (Edwards, n.d)⁸.” Additionally, concerns linger about how the RESILIA tissue (the bovine pericardial tissue that is mounted

on the flexible frame of the valve) will perform over extended periods (Edwards, n.d)⁹. Studies investigating calcification in RESILIA tissue yield mixed results. Some studies suggest lower rates compared to older bioprosthetic materials, while others indicate comparable levels. This inconsistency warrants further investigation and raises questions about the tissue's long-term resistance to calcification, a major contributor to valve dysfunction.



Graphical abstract of how long noncoding RNA (lncRNA) lncTSI attenuated aortic valve calcification. lncTSI reduced aortic valve calcification through negative regulation of the TGF-β/Smad3 pathway and inhibited osteoblast differentiation of human aortic valve interstitial cells (VICs). (PubMed, 10)

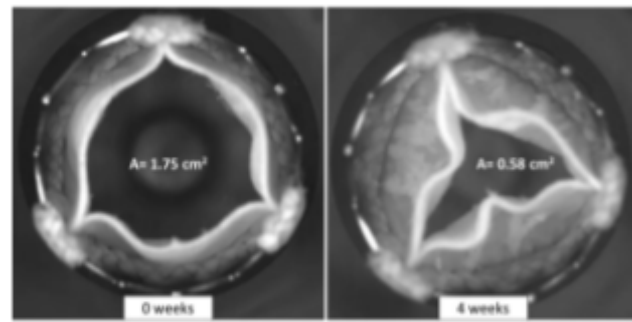
One method to prevent calcification is initial avoidance (PubMed, Liu, Zongtao et al.)¹⁰. Liu, Zongtao, et al. explore the potential of lncTSI, a long non-coding RNA, in mitigating calcific aortic valve disease (CAVD) through its regulatory influence on the transforming growth factor (TGF)-β/Smad3 pathway. The study begins by delineating CAVD as a dynamic process characterized by the osteogenic differentiation of valve interstitial cells (VICs).

Through a series of experiments, the researchers found that lncTSI expression is upregulated in VICs treated with TGF-β1, and its

transcription is modulated by Smad3. Silencing lncTSI intensified TGF-β1-induced Smad3 phosphorylation, subsequently upregulating the expression of key fibrotic genes like RUNX2 and collagen I, while lncTSI overexpression led to their downregulation. Furthermore, in a mouse model of CAVD induced by a high-cholesterol diet, intravenous injection of AAV2-lncTSI significantly mitigated calcium deposition, and reduced expression levels of RUNX2, pSmad3, and collagen I in aortic valve leaflets, consequently resulting in less aortic valve stenosis. These findings suggest that lncTSI might serve as a potential therapeutic target in impeding the progression of CAVD, offering insights into novel diagnostic and therapeutic strategies for this prevalent cardiovascular condition.

In addition to this, the issue of calcification of artificial heart valves is also being investigated by Fraunhofer Institute for Ceramic Technologies and Systems IKTS. They have proposed using a new chemical treatment to prevent the formation of calcium deposits. Calcium deposits build up over the years

with artificial heart valves and can inhibit the function after 10-15 years. The treatment proposed by the institute works by pretreating the valves with glutaraldehyde. This chemical works by counteracting the long-term effects of calcification and stopping calcium ions from binding to the pericardial tissue. This in essence will stop calcification from occurring at all if proven successful. (Fraunhofer)⁷.



Calcification impairs heart valve function. This is the result of an in-vitro calcification test. (Fraunhofer)⁷

Description of product FDA regulatory pathway

For safety and efficiency, it is essential to rely on the United States Food and Drug Administration (FDA) and follow its regulations. The first step is to classify the device. The Edwards Heart Valve Replacement is classified as a Class III medical device, which is critical for life-sustaining purposes and plays a vital role in preventing impairment of human health. It has been assigned the

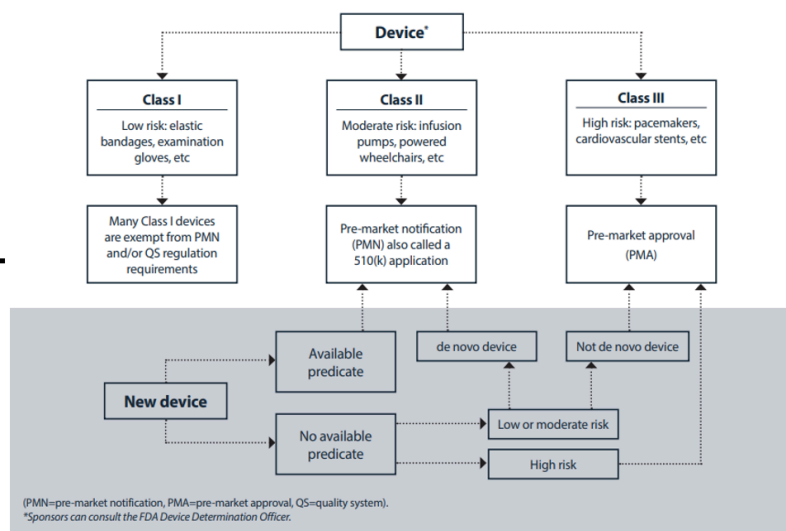


Figure 6: FDA classification and regulatory pathway (Purnama et al., 2018)⁴

product code LWQ. Class III medical devices are associated with the highest risk of illness or injury to patients and require a Premarket Approval Application (PMA), the most rigorous and stringent approval process. During the PMA process, scientific evidence demonstrating the safety and effectiveness of the device

must be provided. Furthermore, the clinical data must be independent, meaning it cannot rely on data supporting another device.

An Investigational Device Exemption (IDE) allows an unapproved device to be used in a clinical study to collect the necessary data to support a PMA submission(*Purnama et al., 2018*)⁴. Therefore, obtaining an IDE is a prerequisite before commencing a clinical study.

The device will undergo a 180-day PMA review and will receive a PMA number at the end of the process. Additionally, the device may need to be registered and listed. The FDA requires all medical device manufacturers to register their facilities and list their devices with the agency (*Purnama et al., 2018*)⁴. As the final step, all medical devices must submit an electronic establishment registration, which will be verified annually.

Conclusion

The field of prosthetic heart valves confronts challenges related to long-term durability and calcification or the necessity for lifelong blood-thinning medications. While companies such as Edwards Lifesciences and Medtronic have contributed significantly to the innovation with numerous patents filed, concerns remain about the long-term performances of prosthetic heart valves. Despite Edwards Lifesciences' use of the ThermaFix process, limitations in long-term durability still exist, making it necessary to further investigate the ability of RESILIA tissue to resist calcification. Methods such as the use of the long non-coding RNA lncTSI as a potential therapeutic target for blocking CAVD and the approach to chemotherapy proposed by the Fraunhofer Institute, if successful, will lead to unprecedented advances in avoiding initial calcification of heart valves. The U.S. Food and Drug Administration's (FDA) regulatory pathway for heart valve replacement emphasizes the importance of rigorous evaluation and adherence to safety protocols, underscores the critical role of clinical data in obtaining Class III medical device approval, and safeguards the safety of these studies. Advances in prosthetic heart valve

technology represent a beacon of hope for patients with heart valve disease, ultimately leading to better outcomes and a higher quality of life for those affected by these challenging conditions.

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